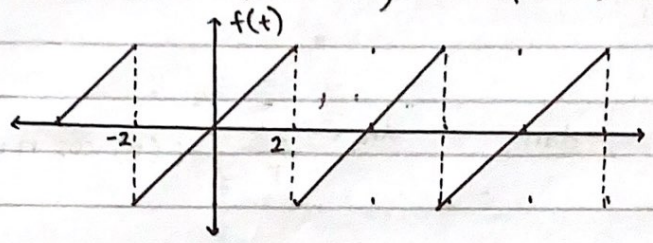
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Latihan 3 !

8. Fungsi $f(t)$ seperti ditunjukkan pada gambar 14 berikut !



Gambar 14 Grafik Fungsi $f(t)$

⇒ Termasuk fungsi ganjil dimana $a_0 = 0$ dan $a_n = 0$

↳ $f(t) = t \quad ; \quad 0 < t < 2$

Periode fungsi $f(t)$ adalah $T = 4$ dan $w = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$

Mencari b_n :

$$b_n = \frac{4}{T} \int_0^{T/2} f(t) \sin nwt \cdot dt$$

$$= \frac{4}{4} \int_0^{2} t \cdot \sin nwt \cdot dt$$

$$= \left[-\frac{1}{nw} t \cos(nwt) + \frac{1}{n^2 w^2} \sin(nwt) \right]_0^2$$

$$= -\frac{2}{nw} \cos(2nw) + \frac{1}{n^2 w^2} \sin(2nw)$$

$$b_1 = -\frac{2}{2 \cdot \frac{\pi}{2}} \cos 2 \cdot 2\frac{\pi}{2} + \frac{1}{2^2 \frac{\pi^2}{2}} \sin\left(2 \cdot \frac{\pi}{2}\right) = -\frac{4}{\pi}(-1) + \frac{2}{\pi^2}(0) = \frac{4}{\pi}$$

$$b_2 = -\frac{2}{2 \cdot \frac{\pi}{2}} \cos 4\frac{\pi}{2} + \frac{1}{4\frac{\pi^2}{4}} \sin 2 \cdot 2 \cdot \frac{\pi}{2} = -\frac{2}{\pi}(1) + \frac{1}{\pi^2}(0) = -\frac{2}{\pi}$$

membentuk pola b_n $\begin{cases} \frac{2}{nw} & \text{bila } n \text{ ganjil} \\ -\frac{2}{nw} & \text{bila } n \text{ genap} \end{cases}$

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∴ Deret Fouriernya

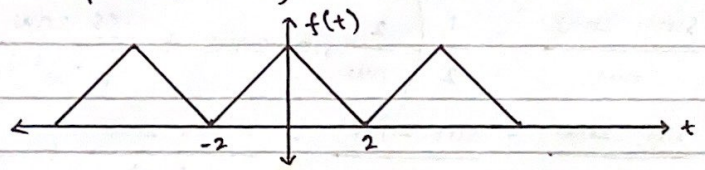
$$f(t) = \sum_{n=1}^{\infty} bn \sin n\omega t$$

$$f(t) = \frac{4}{\pi}\sin\left(\frac{\pi}{2}t\right) - \frac{2}{\pi}\sin\left(\frac{2\pi}{2}t\right) + \frac{4}{3\pi}\sin\left(\frac{3\pi}{2}t\right) - \frac{4}{\pi}\sin\left(\frac{4\pi}{2}t\right) + \frac{4}{5\pi}\sin\left(\frac{5\pi}{2}t\right)$$

atau dapat disederhanakan dengan faktorisasi

$$f(t) = \frac{4}{\pi}\left(\sin\left(\frac{\pi}{2}t\right) - \frac{1}{2}\sin\left(\frac{2\pi}{2}t\right) + \frac{1}{3}\sin\left(\frac{3\pi}{2}t\right) - \frac{1}{4}\sin\left(\frac{4\pi}{2}t\right) + \frac{1}{5}\sin\left(\frac{5\pi}{2}t\right)\right)$$

10. Tentukan deret fourier dari gambar berikut !



↳ f(t) dari grafik di atas merupakan fungsi genap, karena secara grafik f(t) simetris terhadap sumbu Y.

Fungsi genap disebut juga deret cosinus, dimana bn = 0.

$$f(t)\begin{cases} \frac{t}{2}+1 & -2 < t < 0 \\ 1-\frac{t}{2} & 0 < t < 2 \end{cases}$$

Periode fungsi f(t) adalah T = 4 dan $\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$

·) Mencari a_0 :

$$a_0 = \frac{2}{T}\int_0^{T/2} f(t)\,dt \quad \leftarrow a_0 \text{ untuk fungsi genap}$$

$$= \frac{2}{4}\int_0^{2}\left(1-\frac{t}{2}\right)dt$$

$$= \frac{1}{2}\left[t - \frac{t^2}{4}\right]_0^2$$

$$= \frac{1}{2}(1-0)$$

$$= \frac{1}{2}$$

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•) Mencari an :

$$a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega t \cdot dt$$

$$= \frac{4}{4}\int_0^{2}\left(1-\frac{t}{2}\right)\cos n\omega t \cdot dt$$

$$= \int_0^{2}\cos n\omega t - \frac{t}{2}\cos n\omega t \cdot dt$$

$$= \int_0^{2}\cos n\omega t \cdot dt - \frac{1}{2}\int_0^{2} t \cdot \cos n\omega t \cdot dt$$

$$= \left[\frac{\sin n\omega t}{n\omega}\right]_0^{2} - \frac{1}{2}\left[\frac{t}{n\omega}\sin \omega t + \frac{1}{n^2\omega^2}\cos n\omega t\right]_0^{2}$$

$$= \frac{\sin 2n\omega}{n\omega} - \frac{1}{2}\left(\frac{2}{n\omega}\sin 2n\omega + \frac{\cos 2n\omega}{n^2\omega^2} - \frac{1}{n^2\omega^2}\right)$$

$$= \frac{\sin 2n\omega}{n\omega} - \frac{\sin 2n\omega}{n\omega} - \frac{1\cos 2n\omega}{2n^2\omega^2} + \frac{1}{2n^2\omega^2}$$

$$= -\frac{\cos 2n\omega + 1}{2n^2\omega^2}$$

$a_1 = \dfrac{-(-1)+1}{\frac{\pi^2}{2}} = \dfrac{4}{\pi^2}$ → saat an dimana n genap maka hasilnya akan 0.

$a_2 = \dfrac{-1+1}{2\pi^2} = 0$

$a_3 = \dfrac{-(-1)+1}{18\frac{\pi^2}{4}} = \dfrac{4}{9\pi^2}$

$a_4 = 0$

$a_5 = \dfrac{-(-1)+1}{2\cdot 25\cdot\frac{\pi^2}{4}} = \dfrac{4}{25\pi^2}$

$a_7 = \dfrac{4}{49\pi^2}$

$a_9 = \dfrac{4}{81\pi^2}$

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∴ Deret fouriernya

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos nwt$$

$$f(t) = \frac{1}{2} + \frac{4}{\pi^2}\cos wt + \frac{4}{9\pi^2}\cos 3wt + \frac{4}{25\pi^2}\cos 5wt + \frac{4}{49\pi^2}\cos 7wt + \frac{4}{81\pi^2}\cos 9wt$$

$$f(t) = \frac{1}{2} + \frac{4}{\pi^2}\cos\left(\frac{\pi}{2}t\right) + \frac{4}{9\pi^2}\cos\left(\frac{3\pi}{2}t\right) + \frac{4}{25\pi^2}\cos\left(\frac{5\pi}{2}t\right) + \frac{4}{49\pi^2}\cos\left(\frac{7\pi}{2}t\right) + \frac{4}{81\pi^2}\cos\left(\frac{9\pi t}{2}\right)t$$

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